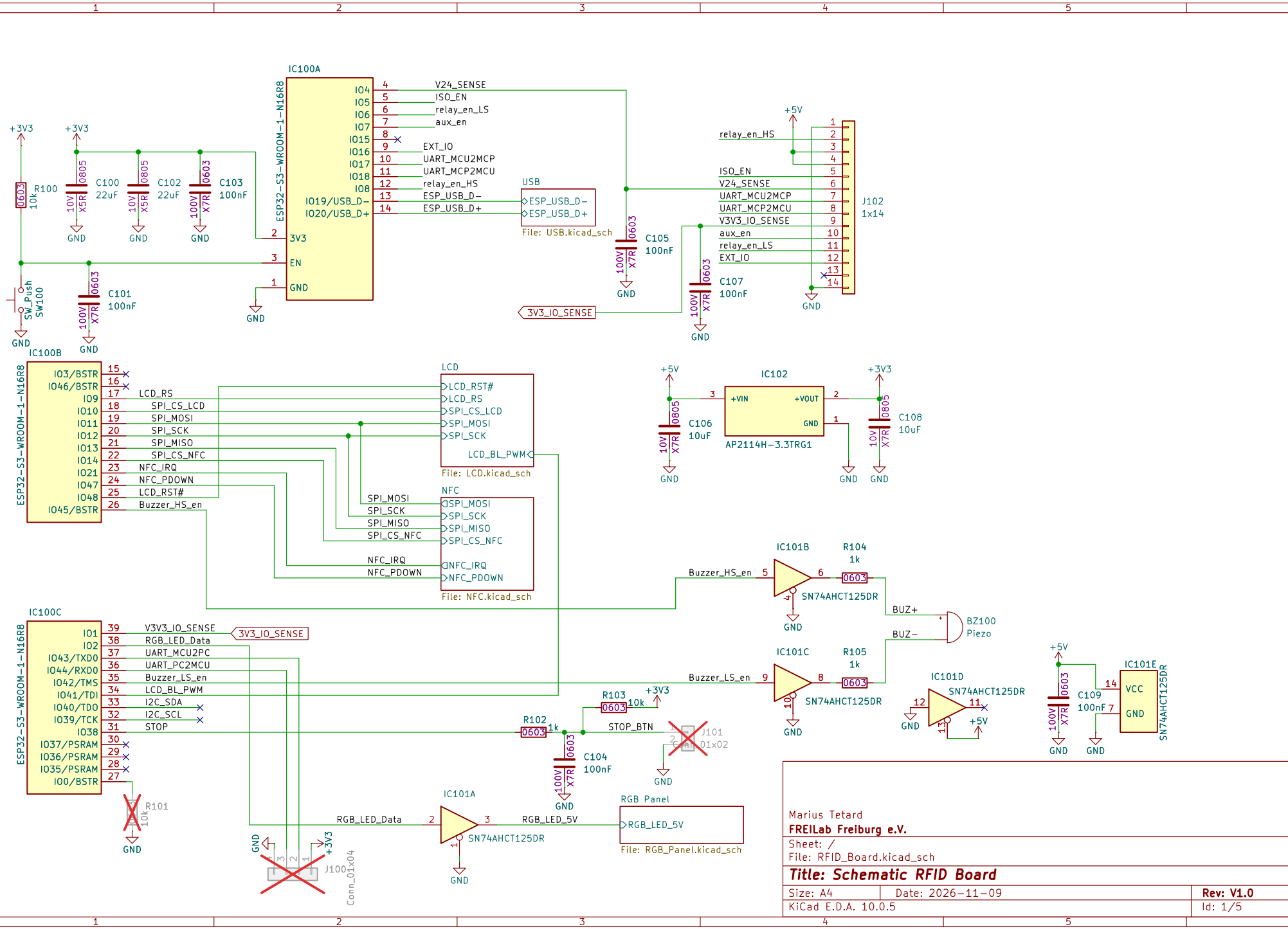
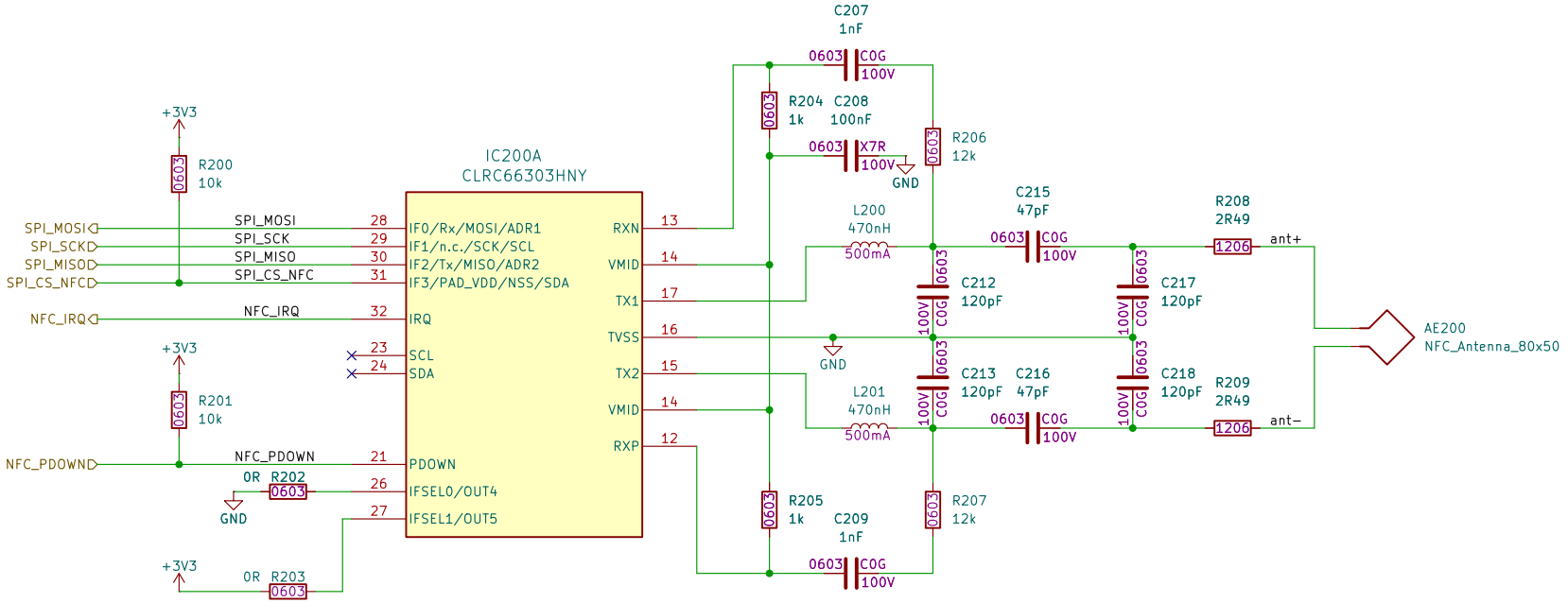
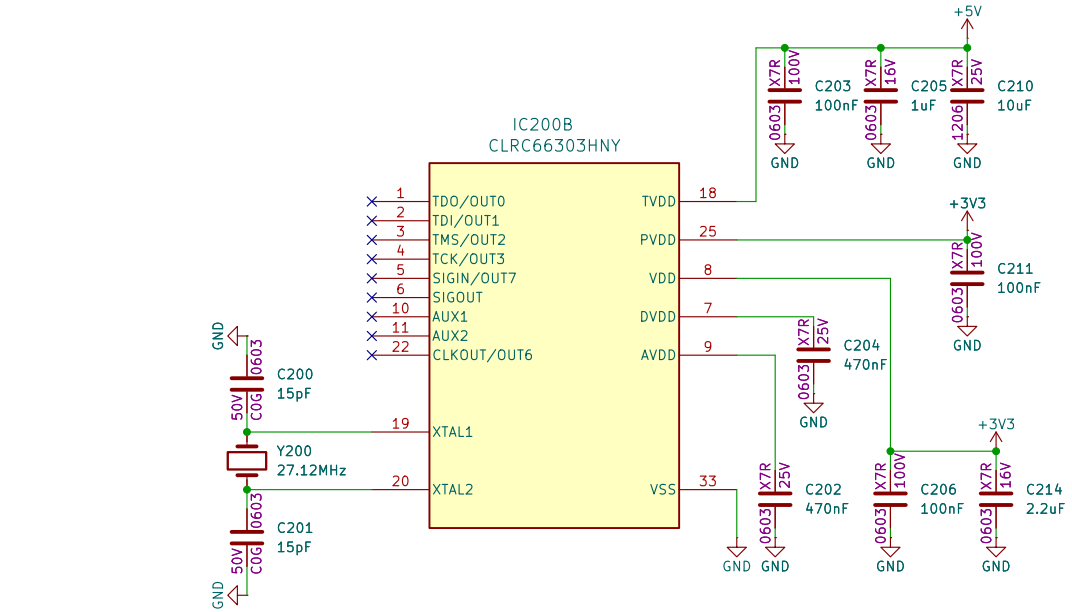
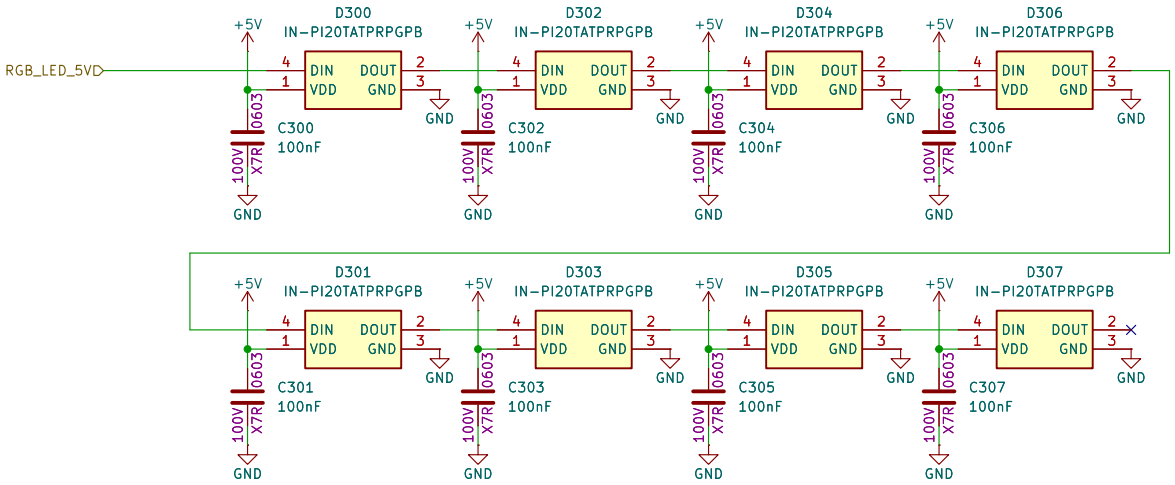

IC100A
ESP32-S3-WROOM-1-N16R8
IO4
IO5
IO6
IO7
IO15
IO16
IO17
IO18
IO8
IO19/USB_D-
IO20/USB_D+
3V3
EN
GND
V24_SENSE
ISO_EN
relay_en_LS
aux_en
EXT_IO
UART_MCU2MCP
UART_MCP2MCU
relay_en_HS
ESP_USB_D-
ESP_USB_D+
USB
File: USB.kicad_sch
+3V3
R100
10k
C100
22uF
C102
22uF
C103
100nF
C101
100nF
SW100
SW_Push
C105
100nF
C107
100nF
3V3_IO_SENSE
+5V
J102
1x14
relay_en_HS
ISO_EN
V24_SENSE
UART_MCU2MCP
UART_MCP2MCU
V3V3_IO_SENSE
aux_en
relay_en_LS
EXT_IO
IC100B
IO3/BSTR
IO46/BSTR
IO9
IO10
IO11
IO12
IO13
IO14
IO21
IO47
IO48
IO45/BSTR
LCD_RS
SPI_CS_LCD
SPI_MOSI
SPI_SCK
SPI_MISO
SPI_CS_NFC
NFC_IRQ
NFC_PDOWN
LCD_RST#
Buzzer_HS_en
LCD
LCD_RST#
LCD_RS
SPI_CS_LCD
SPI_MOSI
SPI_SCK
LCD_BL_PWM
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NFC
SPI_MOSI
SPI_SCK
SPI_MISO
SPI_CS_NFC
NFC_IRQ
NFC_PDOWN
File: NFC.kicad_sch
IC102
+VIN
+VOUT
GND
AP2114H-3.3TRG1
C106
10uF
C108
10uF
IC100C
IO1
IO2
IO43/TXD0
IO44/RXD0
IO42/TMS
IO41/TDI
IO40/TDO
IO39/TCK
IO38
IO37/PSRAM
IO36/PSRAM
IO35/PSRAM
IO0/BSTR
V3V3_IO_SENSE
RGB_LED_Data
UART_MCU2PC
UART_PC2MCU
Buzzer_LS_en
LCD_BL_PWM
I2C_SDA
I2C_SCL
STOP
R101
10k
IC101B
R104
1k
SN74AHCT125DR
BUZ+
BUZ-
BZ100
Piezo
IC101C
R105
1k
SN74AHCT125DR
IC101D
SN74AHCT125DR
IC101E
SN74AHCT125DR
C109
100nF
R102
1k
R103
10k
STOP_BTN
J101
01x02
C104
100nF
IC101A
SN74AHCT125DR
RGB_LED_Data
RGB_LED_5V
RGB Panel
File: RGB_Panel.kicad_sch
J100
Conn_01x04
Marius Tetard
FREILab Freiburg e.V.
Sheet: /
File: RFID_Board.kicad_sch
Title: Schematic RFID Board
Size: A4
Date: 2026-11-09
Rev: V1.0
KiCad E.D.A. 10.0.5
Id: 1/5





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FREILab Freiburg e.V.	
Sheet: /NFC/	
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Title: Schematic RFID Board – NFC	
Size: A4	Date: 2026-11-09
KiCad E.D.A. 10.0.5	Id: 2/5

Rev: V1.0



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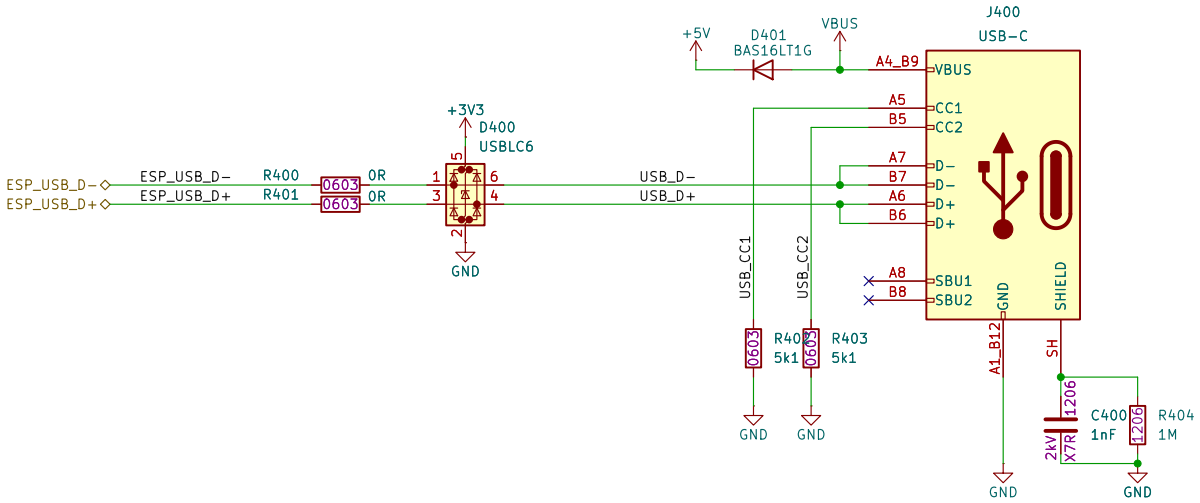
FREILab Freiburg e.V.

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Title: Schematic RFID Board – RGB Panel

Size: A4	Date: 2026-11-09	**Rev: V1.0**
KiCad E.D.A. 10.0.5		Id: 3/5



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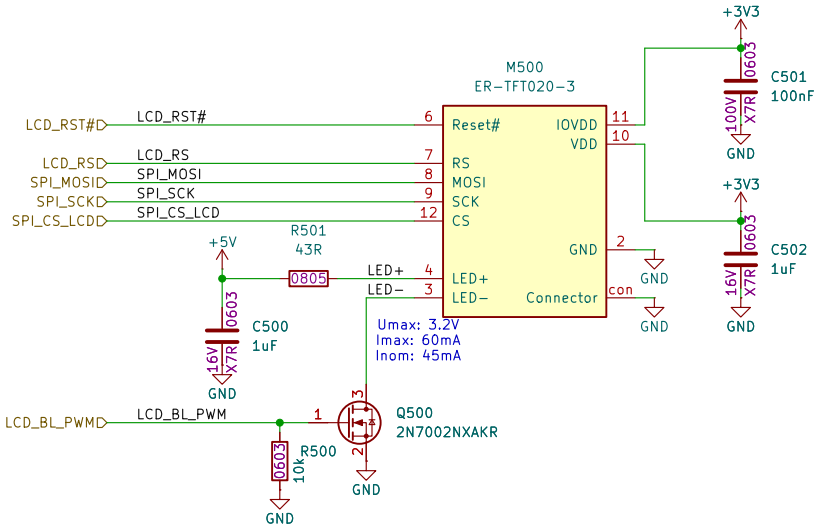
FREILab Freiburg e.V.

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Title: Schematic RFID Board – USB

Size: A4	Date: 2026-11-09	**Rev: V1.0**
KiCad E.D.A. 10.0.5		Id: 4/5



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Sheet: /LCD/

File: LCD.kicad_sch

Title: Schematic RFID Board – LCD

Size: A4	Date: 2026-11-09	**Rev: V1.0**
KiCad E.D.A. 10.0.5		Id: 5/5